

Final Project: Build and Deploy Your Own AI System

Overview

For your final project, you will design, develop, and deploy a functional AI system that reflects a domain of your choosing — whether that's finance, marketing, operations, healthcare management, retail, real estate, or any other area relevant to your studies and interests. This is an individual project that asks you to move beyond passive understanding of AI and machine learning concepts and actually build something that works in the real world.

At its core, the project has two components: (i) building an AI model and (ii) making it interactively accessible to others through a web interface. By the end of the course, you will have a live, interactive application that anyone can visit in a browser, upload their own data, and receive predictions or insights generated by your model.

What You Will Build

Core Track — Prediction Model with Web Deployment

You will develop a supervised machine learning model trained on a dataset relevant to their domain of interest. Examples might include a model that predicts customer churn for a subscription business, estimates housing prices based on property characteristics, forecasts monthly sales revenue, classifies loan applicants by credit risk, or identifies which marketing campaigns are likely to convert. The choice is yours, and you are encouraged to pursue something you genuinely find interesting.

Once your model is trained and validated, you will save it locally (e.g., using tools like pickle or joblib in Python) and integrate it into a web application. The web interface should allow a user to upload a data file or fill in input fields, submit that data to your model, and receive predictions displayed clearly on the page. The experience should be clean, intuitive, and functional — your classmates and instructor will be using it.

Advanced Track — Interactive AI System with LLMs

If you are interested in building something more ambitious, you may extend beyond a traditional prediction model and incorporate large language models (LLMs) as a component of your system. For example, you might build an application that combines a predictive model with a conversational interface, allowing users to ask follow-up questions about their results. Alternatively, you could develop an LLM-powered tool that analyzes unstructured text data — such as customer reviews, financial filings, or support tickets — and produces structured insights. Systems on this track should still have a clear, domain-grounded use case and a functional web interface.

Technical Expectations

You are expected to handle the full stack of this project independently: data sourcing and preprocessing, model training and evaluation, saving the trained model, building the backend logic that loads and runs the model on new inputs, and deploying a working web interface. There is no prescribed technology stack, but common approaches include Python with scikit-learn for modeling, Flask or FastAPI for the backend, and Streamlit or basic HTML/CSS/JavaScript for the frontend. You are welcome to use any tools or frameworks you are comfortable with, as long as the final product is accessible via a browser and functions as described.

Deliverables

You will submit two things and participate in one required meeting.

The first deliverable is a working web application, deployed and accessible via a public URL that your instructor can visit without any special setup. The second is a technical report (roughly 5–10 pages) documenting your problem framing, dataset, modeling choices, evaluation results, and design decisions for the web interface. The report should read as a professional document — imagine you are presenting this work to a manager or client who wants to understand what the model does, why you built it this way, and how well it performs.

In addition to these written deliverables, each student will meet individually with the instructor to walk through their application live. Come prepared to demonstrate the full functionality of your system, explain the decisions you made, and answer questions about your modeling approach and design choices. These meetings are not a formal presentation — think of them as a technical conversation about your work.

Following the individual meetings, a selection of projects that demonstrate exceptional system development will be invited to present their work to the full class. These presentations serve as exemplars for the cohort.

After the exemplar presentations, all students will submit a reflection paper (roughly 3–5 pages). In it, you will compare your own system to the exemplars presented, honestly assessing what could have been done better in your work and what you would change given another iteration. If you believe your system is comparable in quality to those presented, you are welcome to make that case — but you must do so with specific, well-reasoned arguments rather than general claims. The reflection is an opportunity to demonstrate critical thinking about AI system design, not simply to evaluate others.

Evaluation Criteria

Your project will be evaluated across four dimensions. The first is problem framing and domain relevance — does the AI system address a meaningful, clearly defined business problem, and does it reflect genuine understanding of the domain? The second is model quality — is the model properly trained, evaluated with appropriate metrics, and free of issues like severe overfitting? The third is web application functionality — does the interface work correctly, handle different inputs gracefully, and provide a usable experience? The fourth is communication — does your report clearly explain your process and findings, and does your demo convey the value of what you built to a non-technical audience?

A Note on Scope

You do not need to build something that would survive production at a Fortune 500 company. What matters is that your system is thoughtful, functional, and grounded in a real business context. A well-executed prediction model on a carefully chosen dataset, paired with a clean and working web interface, is entirely sufficient for an excellent grade. Complexity is not rewarded for its own sake — clarity, quality, and genuine insight are.